Kirolos Sedra

Software Engineer
Waterloo, Ontario | Relocatable | Authorized to work in Canada; no sponsorship required
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Professional Summary

Backend-focused Software Engineer with experience building Java Spring Boot fintech services, cloud-integrated IoT platforms, and production web systems. Strong in REST APIs, SQL, system integration, test automation, CI/CD, debugging, and translating stakeholder requirements into reliable software releases.

Experience

Aquasensing **Sep 2024 – Present**
Software Engineer *Part-time, Startup Environment*

- Developed features for a cloud-integrated BLE–LTE IoT platform using **React**, **Django**, **Node.js**, and real-time data analytics workflows.
- Built an automated test-and-provisioning suite using **Azure CLI** to configure and validate a **70-unit IoT gateway fleet**.
- Led technical discussions with **Bell IoT** on deployment specifications and pricing, and delivered a stakeholder demonstration for **12Energy Ltd.** that contributed to a **50-gateway commercial deployment** with continued post-sale support.
- Served as Product Owner for the IoT system, introducing disciplined **Git**, Azure DevOps, and Jira workflows; coordinating SDK migrations; and applying database safeguards through stored procedures, logging, and rollback procedures.

ChurchOne / LinC **Ongoing**
Product Owner & Solution Architect *Volunteer, AI-Assisted Development*

- Led a five-person digital-transformation initiative and architected a centralized **React**, **TypeScript**, and **Firebase** platform used by **30+ congregation members** for meetings, attendance, people, notes, assignments, posts, forms, surveys, permissions, PDFs, and administrative workflows.
- Directed AI-assisted engineering through **ChatGPT** and **Codex** by defining requirements, prioritizing features, reviewing implementations, debugging failures, preventing regressions, and validating releases through consistently green frontend and backend **GitHub Actions** pipelines.
- Designed and deployed a secure backend email workflow using **Hono**, **Cloudflare Workers**, **Zod**, Firebase ID-token verification, and **Brevo**, with structured validation, error handling, secret management, bilingual English/Arabic invitations, user training, and production support.

Ejada Systems **Sep 2023 – Aug 2024**
Software Engineer *Full-time, Hybrid*

- Developed and maintained **REST APIs** and backend **microservices** for fintech loan products using **Java Spring Boot**, **SQL**, and stored procedures, supporting systems for **Al Rajhi Bank** and **Emkan Governmental Services**.
- Architected and developed a new **BNPL application** for Emkan Saudi Arabia in a two-person core team, owning backend architecture, business logic, system integration, and frontend support using NewGen and JavaScript.
- Improved SQL-query performance by **30%** through indexing and query optimization, and improved API performance by **50%** by removing redundant stored-procedure calls.
- Supported safe delivery through rollback scripts, database logging, system-integration testing, feature-branch workflows, and production release activities while collaborating with a multicultural Saudi-based team.

Selected Technical Projects

Laboratory Inventory and Experiment Management System *React, TypeScript, Firebase*

- Architected and developed a responsive inventory platform with complete **CRUD** workflows for equipment, quantities, serial numbers, locations, health status, lending, movement histories, projects, tags, comments, and photographs.
- Implemented per-serial tracking, experiment-feasibility validation, tag-based search, mobile workflows, and configurable **PDF reports** containing selected equipment, photographs, notes, health information, and optional movement logs.

Mobile Robot Wireless Performance Analysis *Python, Wi-Fi, 5G NSA/SA, iperf3*

- Designed and executed controlled outdoor, indoor, and RF-chamber experiments comparing Wi-Fi with 5G NSA/SA for mobile-robot connectivity, collecting latency, throughput, RSSI, MCS, and SINR measurements across repeated runs.
- Developed Python-based analysis workflows using motion-aligned windows and per-run P50/P99 statistics to isolate trajectory, load, and signal-condition effects and produce publication-quality figures and tables.

Education

University of Waterloo **Sep 2024 – Aug 2026**
Master of Applied Science in Electrical and Computer Engineering *92% (A+)*
Specialization: Pattern Analysis and Machine Intelligence

Ain Shams University **Sep 2018 – Jul 2023**
Bachelor of Science in Computer Engineering *CGPA: 3.12/4.00*

Technical Skills

Programming: Java, Python, C/C++, JavaScript, TypeScript, SQL, Bash, HTML, CSS
Backend & Web: Spring Boot, Django, Node.js, Hono, React, Firebase, REST APIs, Swagger/OpenAPI, microservices
Testing & Quality: GitHub Actions, Vitest, frontend and backend automated testing, API testing, SIT, regression testing, mocked external-service testing
Cloud & DevOps: Azure, Azure CLI, Azure DevOps, Cloudflare Workers, Netlify, Docker, Git, Jira
Systems & Networking: Linux, macOS, TCP/IP, HTTPS, DNS, DHCP, NTP, FTP, SSH
Spoken Languages: English (C1), French (B1)

Leadership & Community

- Served as IEEE AP-S Chapter Webmaster and Treasurer, maintaining chapter web content through **WordPress**.
- Served as Media Lead and Director of Service, mentoring youth, supporting academic instruction for high-school students in orphanages, and participating in weekly community outreach.